

ESAT Physics

SPEED, RECALL & PROBLEM-SOLVING GUIDE

Speed, Recall & Problem-Solving Guide

Revised edition — real typeset formulas, a fixed 3-tier priority system,
and additional speed tricks folded in

Basis and scope

Syllabus content and terminology are based on UAT-UK, ESAT Guide Physics, June 2026. The official guide emphasises understanding as well as recall. This document keeps the same topic structure but targets a knowledgeable student: formula selection, causal chains, graph interpretation, units and fast elimination of common misconceptions. Priority rankings, shortcuts and speed tactics are strategic synthesis. They are not official topic weightings or claims about question frequency.

How to use this guide

Revision note: the priority tiers below are relabelled from the original P1/P2/P3 numbering to AUTOMATE / FLUENT / TRAP-GUARD. The original numbering collided with the syllabus's own unit codes (e.g. Physics unit "P3 Mechanics" vs. priority tag "P1" for that same row) — the new labels mean the same thing but can never be confused with a unit code.

AUTOMATE. Instant recognition, zero hesitation.

FLUENT. A few written steps, but no uncertainty about method.

TRAP-GUARD. Familiar content where avoidable marks are lost.

- Memorise cold. Facts, identities and patterns that should not consume thinking time.
- Speed trick. A valid shortcut or method-selection rule; use it only when its conditions are satisfied.

Overall preparation priorities

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	P3 Mechanics	Largest concentration of linked quantitative tools: graphs, forces, momentum and energy.
AUTOMATE	P6 Waves	Many concepts change together; frequency, speed, wavelength and direction must be tracked precisely.
AUTOMATE	P1 Electricity	Circuit reasoning plus formula fluency; prefix and meter-placement errors are avoidable.
AUTOMATE	P2 Magnetism	Direction rules, induction and transformers reward fast causal reasoning.
FLUENT	P5 Matter	Particle model plus several short formula families: gas law, latent heat, density and pressure.
FLUENT	P7 Radioactivity	Mostly compact recall plus conservation in decay and half-life interpretation.
FLUENT	P4 Thermal physics	Conceptual mechanisms are straightforward but easy to mix; SHC provides the main calculation.

Each unit below follows the same shape: Priority ranking, Memorise cold, Ranked speed tricks (plus any additional tricks folded in for this revision, clearly marked), Common time-wasters and traps, and a 10-second pre-answer check.

P1. Electricity

Official guide: pp. 5-31 | Sections P1.1-P1.2

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Move between the microscopic picture (electrons/charge) and circuit quantities without mixing up current, voltage, resistance, power and energy. The fastest route is usually to identify what is conserved/equal in the circuit before choosing a formula.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Series/parallel reasoning	Current and voltage rules often solve the circuit before any formula is needed.
AUTOMATE	V, I, R, P, E formula selection	Know which two known quantities determine the third; base SI units prevent prefix errors.
AUTOMATE	Meters and V-I graphs	Connection, resistance and graph interpretation are common conceptual traps.
AUTOMATE	Thermistor/LDR/diode circuits	Track the whole-circuit consequence of a component change.
FLUENT	Electrostatics and earthing	Conceptually simple; electron direction and induction must be precise.
FLUENT	AC/DC and conductors	Mostly recall and interpretation.

MEMORISE COLD

- Current $I=Q/t$; charge $Q=It$. $1\text{ A} = 1\text{ C/s}$.
- Voltage $V=E/Q$: energy transferred per unit charge.
- Resistance $R=V/I$.
- Power $P=IV=I^2R$; the guide also derives $P=V^2/R$ as a useful consequence.
- Energy $E=Pt=VIt$.
- Series: same current; voltages add; resistances add.
- Parallel: same voltage across branches; branch currents add to total; combined resistance is less than any individual branch resistance.
- Voltmeter in parallel, very high resistance. Ammeter in series, very low resistance.
- Fixed resistor V-I graph straight through origin at constant temperature; filament lamp resistance rises as it heats. At a point on a non-linear V-I curve, resistance is V/I , not the tangent gradient.
- NTC thermistor: hotter $\rightarrow$ lower resistance. LDR: brighter $\rightarrow$ lower resistance. Ideal diode conducts in one direction only.
- Charging is electron transfer. Lose electrons $\rightarrow$ positive; gain electrons $\rightarrow$ negative. Charge is conserved.

RANKED SPEED TRICKS

1. Start with circuit topology, not formulae. Mark which components share current (series) and which share voltage (parallel).
2. Meter type from connection. Parallel across one component $\rightarrow$ voltmeter; in the current path $\rightarrow$ ammeter. Ignore where the symbol happens to be drawn on the page.
3. Convert prefixes before calculating. $m=10^{-3}$, $\mu=10^{-6}$, $k=10^3$, $M=10^6$. Most numerical mistakes are powers of ten, not physics.

4. Choose the power form that uses what you know. If I and R are given, $P=I^2R$ avoids first finding V ; if V and R are known, the derived V^2/R form is efficient.
5. For a parallel split, lower resistance takes more current. With equal branch voltage, $I=V/R$; branch currents are inversely proportional to resistance.
6. Component resistance changes in series: track total current first. Example: NTC cools $\rightarrow R$ rises $\rightarrow$ total R rises $\rightarrow$ current falls; then recalculate how the fixed supply voltage is shared.
7. Filament graph: distinguish V -versus- I from I -versus- V . The apparent curvature direction depends on axes, but the physical fact is resistance rises with temperature.
8. Electrostatic induction: ask what electrons do. Positive nearby charge attracts electrons; negative nearby charge repels them. Earthing lets electrons enter/leave.
9. Earthing prevents dangerous potential build-up. Think of a conducting path that allows excess charge to flow instead of accumulating until a spark occurs.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Using the gradient of a curved V - I graph as resistance.
- Putting an ammeter in parallel or a voltmeter in series.
- Mixing mA/microA/kV/Mohm with base units.
- Saying positive charge moves between rubbed solids; in this model electrons move.
- Assuming parallel resistors add like series resistors.
- Forgetting that voltage is energy per charge, not rate of charge flow.

UNDER TIME PRESSURE: 10-SECOND CHECK

- Series or parallel?
- What is equal/conserved?
- Base units?
- Which P formula is shortest?
- Meter by connection?
- Non-ohmic: V/I not curve gradient?
- Electron movement?

EXPANDED PATTERN BANK

P1. Electricity - circuit reasoning patterns

The quickest circuit solutions start with topology: what is the same, what adds, and what changes when a component changes.

P1: MARK THESE BEFORE CALCULATING

- Series path: same current through every component; voltage drops add; resistances add.
- Parallel branches: same voltage across each branch; branch currents add to total; lower resistance branch carries more current because $I=V/R$.
- Ammeter: in series, very low resistance. Voltmeter: in parallel, very high resistance.
- Fixed supply voltage: if one series resistance rises, total resistance rises and current falls; the voltage division between components then changes.

Formula-selection shortcuts

$Q=It$; $V=E/Q$; $R=V/I$; $P=IV=I^2R=V^2/R$; $E=Pt=VIt$. Choose the form that uses quantities already known. Convert mA, microA, kOhm and MOhm before arithmetic. A large fraction of numerical errors are prefix errors, not physics errors.

Component-change chains

NTC thermistor cools: R rises. In series $\rightarrow$ total R rises $\rightarrow$ current falls. With a fixed resistor, its voltage IR falls, so a larger share of the fixed supply voltage appears across the thermistor.

LDR gets brighter: R falls; reverse the same chain.

Filament lamp: higher current heats filament $\rightarrow$ resistance rises. On a non-linear V - I curve use $R=V/I$ at the point, not the tangent gradient.

CONCEPTUAL TRAPS

Current is rate of charge flow; voltage is energy transferred per unit charge. They are not two names for "amount of electricity". Conventional current direction is opposite electron drift in metals. A charged neutral object attracted to another charged object may be responding by induction; attraction alone does not identify sign.

10-SECOND CHECK

Series or parallel? What is equal? What adds? Meter connection? Prefixes in SI? Which P formula avoids an extra step? If resistance changes, did I update total current before voltage sharing?

Expansion page: official syllabus/content basis retained; ranking and speed tactics are strategic synthesis.

P2. Magnetism

Official guide: pp. 32-81 | Sections P2.1-P2.5

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Keep five ideas separate but connected: permanent magnet fields, fields from currents, force on a current, induction from changing field, and transformer ratios. Most mistakes come from using the wrong direction rule or confusing cause with effect.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Motor effect and direction	$F=BIL$ plus Fleming left-hand rule; distinguish force from field direction.
AUTOMATE	Electromagnetic induction	Magnitude and direction depend on rate and sense of field change.
AUTOMATE	Transformers and transmission	Ratios and power conservation give fast quantitative questions.
AUTOMATE	Fields from currents	Right-hand grip and solenoid polarity must be automatic.
FLUENT	Permanent/induced magnetism	Soft/hard material choice and field-line interpretation are mostly conceptual.
FLUENT	DC motor / AC generator structures	Track commutator/slip rings and how the output/turning direction is maintained.

MEMORISE COLD

- Magnetic field lines outside a magnet go N → S; they do not cross; closer spacing means stronger field.
- Soft magnetic material (iron): magnetises/demagnetises easily; good electromagnet core. Hard magnetic material (steel): retains magnetisation; good permanent magnet.
- Straight current-carrying wire: concentric circular field; right-hand grip gives direction. Solenoid field resembles a bar magnet and is strong/approximately uniform inside.
- Motor effect: $F=BIL$ for a straight wire at right angles to a uniform field. Fleming left hand: First finger field, seCond finger conventional current, thuMb force/motion.
- Induced voltage occurs when a conductor cuts field lines or magnetic field through a circuit changes. Faster change and more turns → larger induced voltage.
- Direction of induced effect opposes the change that caused it (the guide explains this using energy conservation/Lenz-type reasoning).
- Transformer: $V_p/V_s = n_p/n_s$. For ideal 100% efficiency, $V_p I_p = V_s I_s$. High-voltage transmission lowers current for the same power and therefore reduces $I^2 R$ cable losses.

RANKED SPEED TRICKS

1. Name the phenomenon before using a hand rule. Right-hand grip = field produced by current. Fleming left hand = force on a current in an external field.
2. Solenoid end shortcut. Viewed end-on: anticlockwise conventional current gives a north pole; clockwise gives south.
3. Motor force magnitude: ask only B, I, L. For the specified right-angle case, double any one doubles F.
4. Induction direction: oppose the change. Approaching north pole → near end of coil becomes north to repel; receding north → near end becomes south to attract.

5. Generator output: change sign every half-turn. The cutting direction reverses, producing alternating voltage.
6. DC motor: current in the coil must reverse each half-turn. The split-ring commutator maintains torque in the same rotational sense.
7. Transformer ratio directly before power algebra. Turns ratio gives voltage ratio immediately; if ideal, use $V_p I_p = V_s I_s$ for the current.
8. Power transmission: hold transmitted power in mind. $P = VI$ means higher $V \rightarrow$ lower I ; loss $I^2 R$ falls with the square of current.
9. Permanent magnet test: repulsion is decisive. Attraction alone could be induced magnetism in an unmagnetised magnetic material.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Using Fleming left hand to find the field around a wire.
- Confusing electron flow with conventional current in hand rules.
- Thinking a transformer works from steady DC; induction requires changing flux.
- Forgetting current ratio is inverse to voltage/turns ratio in an ideal transformer.
- Calling iron hard magnetic because it is mechanically hard/soft; magnetic hardness is about retaining magnetisation.
- Assuming attraction proves an object is permanently magnetised.

UNDER TIME PRESSURE: 10-SECOND CHECK

- Field from current or force on current?
- Which hand rule?
- Induction: what change is being opposed?
- Motor commutator / generator slip rings?
- Turns ratio first?
- Transmission loss $I^2 R$?

EXPANDED PATTERN BANK

P2. Magnetism - field, force and induction decision tree

Keep three questions separate: what creates the field, what force the field causes, and what changing field induces.

WHICH RULE?

- Field around a current: right-hand grip rule. Straight wire $\rightarrow$ concentric circles; solenoid $\rightarrow$ bar-magnet-like field, approximately uniform inside.
- Force on a current in an external field: Fleming left-hand rule. Use conventional current.
- Induced voltage/current: changing magnetic flux or conductor cutting field lines. The induced effect opposes the change that produced it (Lenz-style reasoning).

Cause $\rightarrow$ effect chains

Motor: current in magnetic field $\rightarrow$ force on opposite sides of coil $\rightarrow$ turning effect. Split-ring commutator reverses coil current every half-turn so the turning direction continues.

Generator: changing flux through coil $\rightarrow$ induced voltage. Slip rings allow alternating output from an AC generator.

Transformer: changing primary current $\rightarrow$ changing magnetic field/flux $\rightarrow$ induced secondary voltage. Steady DC gives no continuing secondary voltage because the flux is not changing.

Transformer and transmission shortcuts

Ideal transformer: $V_p/V_s = N_p/N_s$ and $V_{plp} = V_{sls}$. Therefore a voltage step-up is a current step-down. For transmitted power $P = IV$, raising V reduces I ; cable heating loss is I^2R , so reducing current strongly reduces loss.

FAST MISCONCEPTION FILTER

Attraction does not prove permanent magnetism; repulsion does. Soft/hard magnetic refers to ease of magnetisation/demagnetisation, not mechanical hardness. Field lines outside a magnet run N to S and never cross. Do not use Fleming left hand for the field around a wire.

10-SECOND CHECK

Field from current, force on current, or induction? Which hand rule? What change is opposed? Motor commutator or generator slip rings? Transformer needs AC/change? Turns ratio before power?

Expansion page: official syllabus/content basis retained; ranking and speed tactics are strategic synthesis.

P3. Mechanics

Official guide: pp. 82-146 | Sections P3.1-P3.7

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Choose between graphs, forces, momentum and energy before calculating. A knowledgeable student should avoid solving the same motion problem twice: use the quantity that is conserved or the equation that eliminates the unwanted variable.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Kinematics graphs and signs	Gradient/area relationships plus scalar/vector distinctions underpin the unit.
AUTOMATE	Newton laws and force diagrams	Correct objects and force pairs matter more than algebra.
AUTOMATE	Energy method selection	Energy conservation can bypass force/time details.
AUTOMATE	Momentum and impulse	Signed one-dimensional momentum is a fast route through collisions/explosions.
AUTOMATE	Mass, weight, free fall, terminal velocity	Resultant-force reasoning links directly to acceleration.
FLUENT	Springs	Hooke and elastic energy are formulaic but require correct extension and limits.
FLUENT	Work, power, efficiency	Straightforward if energy transfers and units are explicit.

MEMORISE COLD

- Scalars: magnitude only. Vectors: magnitude + direction. Distance/speed are scalar; displacement/velocity/acceleration/force/momentum are vector.
- Speed=distance/time; velocity=displacement/time; acceleration=change in velocity/time; average speed=total distance/total time.
- Graph rules: gradient of displacement-time = velocity; gradient of velocity-time = acceleration; area under velocity-time = displacement; area under speed-time = distance.
- Required equation of motion: $v^2 - u^2 = 2as$ for constant acceleration. The official guide also derives $v = u + at$ and $s = (u+v)t/2$ as useful alternatives for uniform acceleration.
- Hooke: $F = kx$ up to the limit of proportionality. Elastic energy $E = (1/2)Fx = (1/2)kx^2$.
- Newton 1: zero resultant external force $\rightarrow$ rest or uniform straight-line motion. Newton 2: resultant $F = ma$. Newton 3: forces are equal/opposite, same type, simultaneous, and act on different bodies.
- Weight $W = mg$; on Earth use g approximately 10 N/kg unless another value is given. Free fall means weight is the only force.
- Momentum $p = mv$; closed-system momentum conserved; force = rate of change of momentum.
- Work = force x distance moved in the direction of the force. $GPE = mgh$. $KE = (1/2)mv^2$. Power=energy/time. Efficiency=useful output/total input.

RANKED SPEED TRICKS

1. Choose a positive direction before 1-D vector work. Then keep velocity, acceleration, force and momentum signed consistently.

2. Graph first: gradient or area may be the whole question. Do not convert a graph problem into equations unless needed.
3. No time given? Try $v^2 - u^2 = 2as$. It deliberately eliminates t .
4. No time/force details but heights and speeds? Try energy. $KE \leftrightarrow GPE \leftrightarrow$ elastic energy often replaces a multi-step kinematics solution.
5. Collision/explosion with negligible external force? Try momentum before Newton. Write one signed conservation equation.
6. Force diagram: only real forces acting on the chosen object. Do not add a "resultant force" arrow as an extra physical force.
7. Newton 3 pair test: different objects, same interaction. Weight and normal on one object are generally not a third-law pair.
8. Terminal velocity: resultant force zero, not forces absent. Weight downward equals drag upward; acceleration is zero while velocity is non-zero.
9. Airbag/crumple zone logic: same momentum change over longer time $\rightarrow$ smaller force. Use $F = \text{change in momentum} / \text{time}$.
10. Spring energy from force-extension graph. Area under $F-x$ graph is energy; for the linear region it gives $(1/2)Fx$.
11. Apparent weightlessness: look for zero support force. In free fall weight remains, but contact/support force can vanish.
12. Check whether average is speed or velocity. Average speed uses total distance; average velocity uses displacement.

ADDITIONAL SPEED TRICKS — *new, not in the original guide*

1. Terminal velocity shortcut: at terminal velocity the resultant force is zero by definition, so just set weight equal to whatever the drag-force expression is and solve directly — skip any reasoning about acceleration.

OPTIONAL EXTRAS — *useful if already familiar, not a preparation priority*

- Equal-mass elastic collision: the two objects' velocities swap completely. Correct only for this specific one-dimensional, perfectly elastic, equal-mass case — a narrow pattern, worth knowing if already familiar but easy to mis-apply outside that exact setup.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Using distance where displacement is required, or magnitudes where signed vectors are required.
- Using constant-acceleration equations when acceleration is not constant.
- Drawing velocity/acceleration/resultant as though they were extra forces in a force diagram.
- Calling normal contact force simply "reaction" and then pairing it incorrectly under Newton 3.
- Forgetting momentum is signed in one dimension.
- Saying terminal velocity means no forces act.
- Using total path length rather than height difference in mgh .

UNDER TIME PRESSURE: 10-SECOND CHECK

- Positive direction fixed?
- Graph gradient/area?
- Constant acceleration?
- Energy or momentum shortcut?
- Only real forces?
- Newton 3 pair on different bodies?

- Distance vs displacement?
- Units/signs?

EXPANDED PATTERN BANK

P3. Mechanics - choose the method before the equation

A strong student should decide whether the question is fundamentally about a graph, constant acceleration, force, momentum or energy.

METHOD-SELECTION LADDER

- Graph given? Try gradient or area first.
- Constant acceleration and no time? $v^2 - u^2 = 2as$ eliminates t .
- Constant acceleration with u, v, t ? average velocity $= (u+v)/2$, then $s = \text{average velocity} \times t$.
- Forces and mass? draw actual forces, find resultant, then $F=ma$.
- Collision/explosion? signed momentum conservation is usually shortest.
- Heights/speeds/springs with negligible losses? energy conservation may bypass time and force details.

Graph rules to know without derivation

Displacement-time gradient = velocity. Velocity-time gradient = acceleration. Velocity-time signed area = displacement. Speed-time area = distance. A distance-time graph cannot decrease; a displacement-time graph can.

Sign discipline

Choose a positive direction once. Then velocity, acceleration, force and momentum use signed values consistently. Distance, speed, mass and energy do not acquire a direction sign. In a velocity-time graph, area below the axis contributes negative displacement, not negative distance.

MICRO-DRILL: WHICH TOOL?

- Speed after falling through height, no time $\rightarrow$ energy or $v^2 - u^2 = 2as$.
- Two carts collide $\rightarrow$ momentum.
- Stopping force given change in momentum and time $\rightarrow F = \Delta p / \Delta t$.
- Velocity-time triangle $\rightarrow$ area for displacement.
- Spring extension from equilibrium hanging mass $\rightarrow$ weight = elastic force, then $F = kx$.

10-SECOND CHECK

Positive direction fixed? Constant acceleration? Graph area/gradient? Can energy remove time? Can momentum remove force details? Is every vector signed consistently?

Expansion page: official syllabus/content basis retained; ranking and speed tactics are strategic synthesis.

P3. Mechanics - forces, Newton laws and terminal motion

Most conceptual losses come from mixing forces on one object with interaction pairs between two objects.

FORCE-DIAGRAM RULES

Draw only forces acting on the chosen object. Label type and direction. Do not add "resultant force" as an extra arrow; the resultant is the vector sum of the forces already drawn. Keep velocity and acceleration arrows separate from force arrows.

Newton-law distinctions

Newton 1: zero resultant force $\rightarrow$ zero acceleration $\rightarrow$ rest or constant velocity in a straight line.

Newton 2: acceleration is in the direction of the resultant force; $F=ma$ is the constant-mass form of force = rate of change of momentum.

Newton 3: an interaction pair is equal and opposite, same type, simultaneous and acts on different objects. Weight and normal force on the same object are therefore not a third-law pair.

Terminal velocity and apparent weight

Falling object: initially weight exceeds drag $\rightarrow$ downward acceleration. As speed rises, drag rises $\rightarrow$ resultant falls. At terminal velocity drag=weight $\rightarrow$ resultant zero $\rightarrow$ constant speed.

Apparent weight: the support/contact force is what a scale reads. In free fall the object still has weight mg , but support force is zero.

COMMON WRONG TURNS

"Constant velocity means no forces" is false; it means forces sum to zero. "Action and reaction cancel" is misleading because the pair acts on different objects. Mass is inertia and stays the same between planets; weight is gravitational force and changes with g . Free fall means weight is the only force, not zero weight.

10-SECOND CHECK

Which object? Which forces actually act on it? Resultant direction? Newton 3 pair on different bodies? Mass or weight? Terminal motion means zero acceleration, not zero forces?

Expansion page: official syllabus/content basis retained; ranking and speed tactics are strategic synthesis.

P4. Thermal physics

Official guide: pp. 147-167 | Sections P4.1-P4.4

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Classify the transfer mechanism first, then reason about what increases the rate. The three mechanisms should never blur together: conduction uses microscopic transfer through matter, convection is bulk fluid motion, radiation is infrared electromagnetic transfer.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Conduction/convection/radiation distinction	Most conceptual errors are mechanism confusion.
AUTOMATE	Specific heat capacity	Main calculation in the unit; units and temperature change must be consistent.
FLUENT	Radiation surface factors	Absorber/emitter relationships are common reasoning questions.
FLUENT	Convection density chain	Temperature → expansion → density → bulk flow should be explicit.
TRAP-GUARD	Conduction-rate factors	Straightforward qualitative relationships.

MEMORISE COLD

- Specific heat capacity $c = E / (m \Delta T)$, so $E = mc \Delta T$. Units $\text{J kg}^{-1} \text{degreeC}^{-1}$.
- Conduction requires matter; rate rises with greater temperature difference, better thermal conductor, larger cross-sectional/contact area, and shorter conduction distance. Metals conduct well because free electrons transfer energy quickly in addition to lattice vibrations.
- Convection occurs only in fluids: heating usually expands a fluid, lowers density, and the warmer fluid rises while cooler denser fluid replaces it.
- Thermal radiation is infrared electromagnetic radiation and can cross a vacuum.
- Matt/dark surfaces are better absorbers and better emitters than shiny/light surfaces; shiny surfaces are better reflectors.
- Every object above absolute zero emits and absorbs thermal radiation; net heating/cooling depends on the balance.

RANKED SPEED TRICKS

1. Mechanism classification question: ask what is moving. Energy through neighbouring particles → conduction; bulk fluid → convection; EM through space/vacuum → radiation.
2. Convection chain in one sentence. Warmer fluid expands → density decreases → rises → cooler denser fluid sinks/replaces it.
3. Good absorber = good emitter. Use this single paired fact instead of memorising separate surface rules.
4. Specific heat capacity: use temperature change, not final temperature. $\Delta T = |T_{\text{final}} - T_{\text{initial}}|$ for energy magnitude.
5. Metal feels colder than wood at same temperature because of rate, not lower temperature. Metal conducts energy from the hand faster.
6. Insulation: identify which mechanism is being suppressed. Trapped small air pockets reduce convection and air itself conducts poorly; shiny foil reduces radiation.

7. Constant temperature despite energy transfer signals state change. That links into P5 latent heat rather than SHC heating.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Saying convection occurs in solids.
- Saying radiation needs particles/air.
- Confusing heat capacity of one object with specific heat capacity per kg.
- Using final temperature rather than temperature change in $E=mc \Delta T$.
- Assuming a shiny surface is a good emitter because it looks bright; it is a poor IR emitter/absorber.
- Saying particles themselves expand when a fluid expands.

UNDER TIME PRESSURE: 10-SECOND CHECK

- Conduction, convection or radiation?
- Needs matter?
- Bulk fluid motion?
- Surface matt/shiny?
- ΔT ?
- State change instead of temperature rise?

EXPANDED PATTERN BANK

P4-P5. Thermal physics and matter - distinction bank

These chapters are compact if you classify the mechanism or state change before choosing a formula.

HEAT TRANSFER: CLASSIFY FIRST

- Conduction: energy transfer through matter by particle interactions; metals also use free electrons. Better conductor, larger area, larger temperature difference and shorter path increase the rate.
- Convection: bulk movement of a fluid driven by density differences; it cannot occur in a solid.
- Radiation: electromagnetic infrared transfer; works through a vacuum. Matt/dark surfaces are better absorbers and emitters than shiny surfaces.

Heating versus state change

Temperature changing within one state: $E=mc \Delta T$.

State changing at constant temperature: $E=mL$. Energy changes particle separation/bonding rather than average kinetic energy/temperature.

A flat section of a heating curve is therefore a state-change signal, not "no energy transfer".

Matter formula triggers

Density: $\rho=m/V$. $1 \text{ g/cm}^3=1000 \text{ kg/m}^3$.

Pressure: $P=F/A$. Small area $\rightarrow$ larger pressure for same perpendicular force.

Hydrostatic: $P=h \rho g$ for pressure due to the liquid; total pressure may also include atmospheric pressure if asked.

Boyle: at constant temperature, $P_1V_1=P_2V_2$. If temperature is not constant, do not apply the relation blindly.

COMMON TRAPS

Calling convection "hot air rising because heat rises" without density reasoning; forgetting radiation works through vacuum; using $E=mc \Delta T$ during a phase change; confusing heat capacity with specific heat capacity; using container shape in hydrostatic pressure; forgetting to subtract a sinker volume in displacement measurements.

10-SECOND CHECK

Conduction, convection or radiation? Temperature change or state change? Boyle conditions satisfied? Density units? Pressure due to liquid only or total?

Expansion page: official syllabus/content basis retained; ranking and speed tactics are strategic synthesis.

P5. Matter

Official guide: pp. 168-196 | Sections P5.1-P5.5

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Connect the particle model to measurable pressure, temperature, density and changes of state. The formulas are short; the important skill is knowing what each model assumption means physically and what must remain constant for a relation such as Boyle's law.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Ideal gas pressure-volume reasoning	PV=constant only for fixed amount of gas at constant temperature.
AUTOMATE	Latent heat/state-change graphs	Energy changes particle arrangement while temperature stays constant during the transition.
AUTOMATE	Density and pressure	Straightforward formulas, frequent unit/area/depth traps.
AUTOMATE	Hydrostatic pressure	Distinguish pressure due to liquid from total absolute pressure.
FLUENT	Particle model explanations	Use spacing, motion and forces rather than macroscopic circular explanations.
FLUENT	Experimental density	Choose displacement method for irregular objects and handle sinkers carefully.

MEMORISE COLD

- Solids: particles close, fixed arrangement, vibrate. Liquids: close but disordered, move past each other. Gases: far apart, random rapid motion, negligible interactions except collisions in ideal model.
- Gas pressure comes from particle collisions with container walls; higher temperature at fixed volume means faster particles and higher pressure.
- Boyle: $P_1V_1=P_2V_2$ for a fixed mass of gas at the same temperature.
- Specific latent heat: $E=mL$. Fusion for melting/freezing; vaporisation for boiling/condensing. Temperature stays constant during a pure-state change.
- Density $\rho=m/V$. $1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$.
- Pressure $P=F/A$. Hydrostatic pressure due to a liquid column: $p=h \rho g$.

RANKED SPEED TRICKS

1. Boyle: test the conditions before multiplying. Same fixed amount of gas and same temperature are essential.
2. Gas compression sanity check. Smaller V at constant T $\rightarrow$ larger P; if your arithmetic gives both smaller, reverse the ratio.
3. State-change graph: flat section means latent heat. Energy changes separation/bonding arrangement rather than average kinetic energy/temperature.
4. Density experiment for irregular object: displacement. Volume gained in measuring cylinder/displacement can equals object volume; if using a sinker, subtract the sinker volume.
5. Pressure: smaller area for same force $\rightarrow$ larger pressure. Use this to sanity-check needles, skis, knives and snowshoes.

6. Hydrostatic pressure: depends on depth, liquid density and g , not vessel shape. Points at the same depth in the same connected liquid have the same hydrostatic pressure.
7. Read wording: "pressure due to liquid" excludes atmosphere. If total pressure is given, subtract atmospheric pressure before using $h \rho g$ when appropriate.
8. Unit direction: cm^3 to m^3 divides by 10^6 ; g/cm^3 to kg/m^3 multiplies by 1000. These two are easy to confuse because density is a quotient.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Using Boyle's law while temperature changes.
- Saying gas particles get bigger when heated.
- Using $E=mc \Delta T$ across a state-change plateau instead of $E=mL$.
- Forgetting to subtract a sinker volume in displacement density measurement.
- Using container shape in hydrostatic pressure.
- Adding atmospheric pressure when the question asks only for liquid pressure, or omitting it when total pressure is requested.

UNDER TIME PRESSURE: 10-SECOND CHECK

- Particle-model explanation?
- Boyle conditions satisfied?
- Heating or state change?
- Density units?
- Pressure area perpendicular?
- Hydrostatic: liquid-only or total?

EXPANDED PATTERN BANK

P4-P5. Thermal physics and matter - distinction bank

These chapters are compact if you classify the mechanism or state change before choosing a formula.

HEAT TRANSFER: CLASSIFY FIRST

- Conduction: energy transfer through matter by particle interactions; metals also use free electrons. Better conductor, larger area, larger temperature difference and shorter path increase the rate.
- Convection: bulk movement of a fluid driven by density differences; it cannot occur in a solid.
- Radiation: electromagnetic infrared transfer; works through a vacuum. Matt/dark surfaces are better absorbers and emitters than shiny surfaces.

Heating versus state change

Temperature changing within one state: $E=mc \Delta T$.

State changing at constant temperature: $E=mL$. Energy changes particle separation/bonding rather than average kinetic energy/temperature.

A flat section of a heating curve is therefore a state-change signal, not "no energy transfer".

Matter formula triggers

Density: $\rho=m/V$. $1 \text{ g/cm}^3=1000 \text{ kg/m}^3$.

Pressure: $P=F/A$. Small area $\rightarrow$ larger pressure for same perpendicular force.

Hydrostatic: $P=h \rho g$ for pressure due to the liquid; total pressure may also include atmospheric pressure if asked.

Boyle: at constant temperature, $P_1V_1=P_2V_2$. If temperature is not constant, do not apply the relation blindly.

COMMON TRAPS

Calling convection "hot air rising because heat rises" without density reasoning; forgetting radiation works through vacuum; using $E=mc \Delta T$ during a phase change; confusing heat capacity with specific heat capacity; using container shape in hydrostatic pressure; forgetting to subtract a sinker volume in displacement measurements.

10-SECOND CHECK

Conduction, convection or radiation? Temperature change or state change? Boyle conditions satisfied? Density units? Pressure due to liquid only or total?

Expansion page: official syllabus/content basis retained; ranking and speed tactics are strategic synthesis.

P6. Waves

Official guide: pp. 197-274 | Sections P6.1-P6.5

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Track what changes and what stays fixed. Most wave questions become short once frequency is treated as the anchor at boundaries and once the normal is drawn before any optical angle is interpreted.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	$v=f\lambda$ and $f=1/T$	Core quantitative relationships across all wave types.
AUTOMATE	Reflection/refraction changes	Frequency remains fixed at a boundary; speed and wavelength change together.
AUTOMATE	Ray diagrams and normals	Angle mistakes are usually reference-line mistakes.
AUTOMATE	EM spectrum order/properties	Order, hazards and applications should be rapid recall.
AUTOMATE	Sound/ultrasound/echo	Medium requirement and round-trip timing are common.
FLUENT	Doppler effect	Approach/recede relationships can be answered qualitatively with wavefront spacing.
FLUENT	Wave vocabulary/transverse vs longitudinal	Foundational recognition; should be automatic.

MEMORISE COLD

- Frequency $f=1/T$. Wave speed $v=\text{distance}/\text{time}$ and $v=f\lambda$.
- Amplitude = maximum displacement from equilibrium. Wavelength = distance between neighbouring points in phase (e.g. peak to peak or compression to compression).
- Transverse oscillation perpendicular to energy transfer; longitudinal oscillation parallel.
- Reflection: angle of incidence = angle of reflection, measured from the normal. Reflection in the same medium changes direction but not speed, frequency or wavelength.
- Refraction: crossing a boundary changes speed; frequency remains unchanged; wavelength changes in the same proportion as speed. Slower medium $\rightarrow$ ray bends towards the normal; faster $\rightarrow$ away.
- Echo/ranging with a return trip: one-way distance $= (1/2)vt$.
- Sound: longitudinal, needs a medium, pitch tracks frequency, loudness tracks amplitude, human hearing about 20 Hz to 20 kHz; ultrasound >20 kHz.
- EM order low f /long $\lambda \rightarrow$ high f /short λ : radio, microwave, infrared, visible red-to-violet, ultraviolet, X-ray, gamma. All EM waves are transverse and travel at $c=3.0 \times 10^8$ m/s in vacuum.

RANKED SPEED TRICKS

1. At a boundary, anchor on frequency. The source fixes f , so if speed falls then wavelength falls; if speed rises then wavelength rises.
2. Draw the normal before reading any angle. Incidence/reflection/refraction angles are measured from the normal, never the surface.
3. Refraction direction from speed. Slower $\rightarrow$ towards normal; faster $\rightarrow$ away; along the normal $\rightarrow$ no change of direction though speed/wavelength still change.
4. Pulse ranging: ask whether the clock measured a round trip. If it returned to the source/detector, halve vt unless the question has already defined one-way time.

5. Oscilloscope: horizontal spacing -> period/frequency; vertical size -> amplitude. Read the graph directly before inserting formulae.
6. Doppler shortcut: approaching squeezes wavefronts -> shorter λ /higher observed f ; receding stretches -> longer λ /lower f .
7. Sound versus EM: vacuum is the discriminator. Sound cannot travel through vacuum; EM can.
8. EM order anchor points. Radio at long-wavelength end, gamma at short; visible red nearer IR and violet nearer UV.
9. Reflection/refraction analogy. The official guide deliberately links water-wave crests and optical rays; use whichever picture makes the change clearer.
10. X-ray versus gamma of the same wavelength: production differs, wave itself does not. The official guide notes EM region boundaries overlap rather than being sharp.

ADDITIONAL SPEED TRICKS — *new, not in the original guide*

1. Inside a medium, use $v = f\lambda$ with that medium's own wave speed v — not $c = f\lambda$. The vacuum wave equation only applies in a vacuum; substituting c for the medium's own (lower) speed is a common, silent error the base guide doesn't explicitly flag.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Changing frequency at refraction.
- Measuring optical angles from the surface.
- Forgetting the factor $1/2$ in echo/sonar/radar round-trip distance.
- Calling sound transverse.
- Saying louder sound has higher frequency; loudness is amplitude, pitch is frequency.
- Reversing the EM spectrum order.
- Saying reflection changes wave speed in the same medium.

UNDER TIME PRESSURE: 10-SECOND CHECK

- f or T ?
- $v = f\lambda$?
- Boundary: f unchanged?
- Normal drawn?
- Toward/away from normal?
- Round trip?
- Sound needs medium?
- EM order?

EXPANDED PATTERN BANK

P6. Waves - boundary and ray reasoning

At a boundary, frequency is the anchor. Speed and wavelength may change; the source frequency does not.

BOUNDARY RULES

- Reflection in the same medium: direction changes; speed, frequency and wavelength do not. Angle of incidence = angle of reflection, measured from the normal.
- Refraction: speed changes at the boundary; frequency stays fixed; wavelength changes in the same proportion as speed because $v = f\lambda$.
- Direction: slower medium -> towards normal; faster medium -> away from normal. Along the normal -> no direction change, although speed and wavelength still change.

Visual checks

Draw the normal first. Most optical angle errors come from measuring from the surface.

Wavefront picture: shorter wavelength in the slower region. If one edge of a wavefront enters first and slows first, the front pivots; this is the physical reason for refraction.

Ray picture: ray is perpendicular to wavefronts and shows energy-transfer direction. Use whichever representation makes the question shorter.

Core quantitative shortcuts

$f=1/T$ and $v=f\lambda$. If f is fixed at a boundary, ratios obey $v_1/v_2=\lambda_1/\lambda_2$. For a pulse that travels out and returns, one-way distance $= (1/2)vt$ unless the question explicitly gives a one-way time.

COMMON WRONG TURNS

Changing frequency at refraction; measuring angles from the surface; saying reflection changes speed in the same medium; forgetting the factor $1/2$ on a round trip; treating the normal itself as the boundary.

10-SECOND CHECK

Normal drawn? Reflection or refraction? f unchanged? Slower or faster medium? wavelength follows speed? Round trip?

Expansion page: official syllabus/content basis retained; ranking and speed tactics are strategic synthesis.

P6. Waves - sound, Doppler and the EM spectrum

Separate wave type, amplitude, frequency and propagation medium before reasoning about applications.

SOUND AND OSCILLOSCOPE

Sound is longitudinal and requires a medium. Pitch tracks frequency; loudness tracks amplitude. Human hearing is approximately 20 Hz to 20 kHz; ultrasound is above 20 kHz. On a trace, horizontal spacing gives period (hence frequency); vertical height gives amplitude.

Doppler shortcut

Approaching source/observer $\rightarrow$ wavefronts are squeezed $\rightarrow$ shorter observed wavelength and higher observed frequency. Receding $\rightarrow$ stretched $\rightarrow$ longer wavelength and lower frequency. This is a qualitative spacing argument; do not confuse it with a change in the wave speed in a fixed medium.

EM spectrum recall

Low frequency / long wavelength $\rightarrow$ radio, microwave, infrared, visible red-to-violet, ultraviolet, X-ray, gamma $\rightarrow$ high frequency / short wavelength. All are transverse and travel at c in vacuum. The boundaries between named bands are not sharp. X-rays and gamma rays of the same wavelength are physically the same kind of EM wave; the naming distinction is associated with their production.

Hazard/application filter

Higher-frequency UV, X-ray and gamma can ionise and damage cells. Lower-frequency bands are primarily associated with heating or communication in the guide. For application questions, first match penetration/interaction to the task, then check whether the proposed wavelength/frequency range is plausible.

MICRO-DRILL

- Pulse returns after 0.40 s in water at 1500 m/s $\rightarrow$ depth 300 m because the measured path is double the depth.
- Ambulance approaches $\rightarrow$ higher pitch, not higher sound speed.
- Louder note $\rightarrow$ larger amplitude, not necessarily higher frequency.

10-SECOND CHECK

Longitudinal or transverse? Needs medium? Pitch or loudness? Doppler approach/recede? EM order? Ionising? One-way or round-trip time?

Expansion page: official syllabus/content basis retained; ranking and speed tactics are strategic synthesis.

P7. Radioactivity

Official guide: pp. 275-312 | Sections P7.1-P7.4

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Use conservation of nucleon number and charge to make nuclear equations mechanical, then keep penetration, ionisation and half-life logically separate. Most questions reward precise qualitative comparisons rather than long computation.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Alpha/beta/gamma properties	Mass, charge, penetration and ionisation determine most application/hazard questions.
AUTOMATE	Nuclear equations	Conserve mass number and atomic number; this is faster than memorising daughter nuclides.
AUTOMATE	Half-life and background correction	Repeated halving and graph reading should be automatic.
FLUENT	Applications/hazards	Match penetration and half-life to the task; internal versus external hazard matters.
FLUENT	Atomic/nuclide notation	Compact recall but essential for decay calculations.
FLUENT	Field deflection	Charge and mass explain direction and amount qualitatively.

MEMORISE COLD

- Atomic number Z =protons; mass number A =protons+neutrons. Neutral atom electrons=protons. Isotopes: same Z , different neutron number/ A .
- Proton relative mass/charge 1,+1; neutron 1,0; electron about 1/2000,-1. Ionisation changes electrons, not protons.
- Alpha = helium nucleus ($A=4, Z=2$): +2 charge, strongest ionising, least penetrating. Beta-minus = electron from nuclear conversion: A unchanged, Z of daughter increases by 1. Gamma = EM wave: no mass/charge, most penetrating, least ionising of the three.
- Decay equations conserve total mass number and total atomic/charge number.
- Half-life = time for undecayed nuclei or corrected count rate to halve. After n half-lives, remaining fraction= $(1/2)^n$.
- Correct source count rate = measured count - background count.
- Alpha/beta deflect in opposite directions in electric/magnetic fields; beta deflects more because of much smaller mass; gamma is undeflected.

RANKED SPEED TRICKS

- Balance A and Z like two independent conservation equations. Alpha: subtract 4 and 2. Beta-minus: A same, daughter Z increases by 1. Gamma: A and Z unchanged.
- Penetration/ionisation are inverse rankings. Alpha most ionising/least penetrating; gamma least ionising/most penetrating; beta intermediate.
- Hazard depends on location. Alpha is very dangerous inside the body because its energy is deposited over a short range; external alpha is stopped by skin. Gamma is a more serious external penetration hazard.
- Half-life arithmetic: count halvings. Powers of two are faster than logarithms for the values used in the guide.

5. Going backwards in time: double per half-life. Do not keep halving when asked for an earlier activity/count.
6. Subtract background before half-life work. Background does not decay with the source and otherwise distorts the ratio.
7. Application choice = penetration + timescale. Tracer needs radiation detectable outside and short enough half-life to limit dose; industrial source generally benefits from longer half-life.
8. Field deflection: sign sets direction, mass affects curvature. Gamma no charge $\rightarrow$ no deflection.
9. Random decay means individual nucleus timing is unpredictable. Half-life describes population statistics, not a countdown for one nucleus.

OPTIONAL EXTRAS — *useful if already familiar, not a preparation priority*

- Fractional half-lives: the remaining fraction is $(1/2)^{t/T}$ for any elapsed time t , not only whole multiples of the half-life T . The "count the halvings" shortcut only works when t is a clean multiple of T — this is the general formula behind it.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Changing mass number in beta-minus decay.
- Saying gamma is a particle with charge.
- Treating half-life as dependent on sample size, temperature, pressure or chemical state.
- Doing half-life on raw measured count without removing background.
- Assuming most penetrating means most ionising.
- Calling alpha harmless because paper stops it, without considering internal contamination.

UNDER TIME PRESSURE: 10-SECOND CHECK

- A and Z conserved?
- Alpha/beta/gamma property?
- Internal or external hazard?
- Background subtracted?
- How many half-lives?
- Earlier $\rightarrow$ double / later $\rightarrow$ halve?
- Application: penetration + half-life?

EXPANDED PATTERN BANK

P7. Radioactivity - fast comparison and application logic

Balance decay with two conservation columns, then reason separately about ionisation, penetration and half-life.

ALPHA / BETA / GAMMA COMPARISON

- Alpha: helium nucleus, $A=4$ $Z=2$, charge $+2$; most ionising, least penetrating.
- Beta-minus: electron from nuclear conversion; tiny mass, charge -1 ; intermediate penetration/ionisation. Daughter A unchanged, Z increases by 1.
- Gamma: electromagnetic wave, no mass or charge; most penetrating and least ionising of the three. A and Z unchanged.

Half-life speed method

Subtract background before analysing source count rate. Then count halvings rather than using logarithms when the numbers are powers of two. After n half-lives, remaining fraction $= (1/2)^n$. Half-life is a property of the nuclide and is not changed by sample size, temperature, pressure or chemical state.

Application logic

Medical tracer: radiation must escape the body for external detection and should minimise dose; the guide favours gamma and a short half-life appropriate to the procedure.

Sterilisation: penetrating radiation must reach through equipment/food; long half-life reduces replacement frequency.

Thickness control: partial penetration is useful so detected count changes with thickness; beta is the standard fit.

Hazard logic flips with location

Alpha outside the body is stopped by skin, but an internal alpha source deposits highly ionising radiation over a short range and is especially damaging. Gamma penetrates from outside and can expose internal tissue; inside the body, much may pass through without interacting, so the ranking is not simply "gamma always most dangerous".

DECAY-EQUATION CHECK

Keep two columns: total mass number A and total atomic number/charge Z . Conserve each independently. This is safer than trying to remember the daughter isotope by name.

10-SECOND CHECK

Background removed? A conserved? Z conserved? Internal or external hazard? Need penetration or ionisation? Half-life suited to the application timescale?

Expansion page: official syllabus/content basis retained; ranking and speed tactics are strategic synthesis.

Final recall sheet

Physics - cross-topic formula and reasoning selector

A final decision page for mixed questions. Identify the physical relationship before searching memory for an equation.

IF THE QUESTION GIVES... THINK...

- Charge and time $\rightarrow I=Q/t$. Voltage/current/resistance $\rightarrow V=IR$. Electrical power $\rightarrow P=IV$, I^2R or V^2/R .
- Motion graph $\rightarrow$ gradient/area. Constant acceleration $\rightarrow$ kinematics. Collision $\rightarrow$ momentum.
- Height/speed/spring $\rightarrow$ energy.
- Heating with $\Delta T \rightarrow E=mc \Delta T$. Phase change $\rightarrow E=mL$. Gas compressed at constant $T \rightarrow PV$ constant.
- Wave f , λ , $v \rightarrow v=f \lambda$. Return pulse $\rightarrow$ check factor $1/2$. Decay count $\rightarrow$ background, then repeated halving.

Unit and magnitude filter

Convert prefixes before substitution. Check the dimensions/units of the final quantity: A, V, ohm, W, J, N, kg, m/s, Pa, Hz. A correct-looking number with the wrong unit family signals a wrong relationship. Estimate order of magnitude before detailed arithmetic when options are widely spaced.

Causal-chain filter

When a variable changes, state the first physical consequence before the final answer. Examples: thermistor R changes $\rightarrow$ total circuit current changes $\rightarrow$ voltage sharing changes; wave speed changes $\rightarrow$ wavelength changes while frequency stays fixed; drag changes $\rightarrow$ resultant force changes $\rightarrow$ acceleration changes; transformer voltage changes $\rightarrow$ current changes for fixed power $\rightarrow I^2R$ loss changes.

THE THREE MOST EXPENSIVE MISTAKES

1. Using a formula outside its conditions (SUVAT without constant acceleration; Boyle without constant temperature).
2. Mixing quantities that sound similar (mass/weight, current/voltage, energy/power, speed/velocity, heat capacity/specific heat capacity).
3. Ignoring the reference definition (optical angle from normal; bearing from north; force diagram for one chosen object).

FINAL 10-SECOND CHECK

What principle is this really testing? What is conserved/equal? Are the formula conditions satisfied? SI prefixes converted? Sign/direction fixed? Does the answer have the right unit, sign and order of magnitude?

Expansion page: official syllabus/content basis retained; ranking and speed tactics are strategic synthesis.

Final high-speed recall sheet

Highest-value habits

- Start with the physical relationship before choosing a formula.
- Convert prefixes and units before arithmetic.
- Series: current same; parallel: voltage same.
- Magnetism: right-hand grip for field, Fleming left hand for motor force, oppose change for induction.
- Mechanics: fix a positive direction; use graph area/gradient before equations.
- Try energy when time/forces are irrelevant; try momentum for isolated collisions.
- Thermal: classify conduction/convection/radiation before explaining.
- Matter: Boyle only at constant temperature; latent heat means constant temperature during state change.
- Waves: frequency stays fixed across a boundary; speed and wavelength change together.
- Draw the normal before optical angles.
- Radioactivity: balance A and Z ; subtract background; count half-lives.
- Use order-of-magnitude and causal-direction checks after every calculation.

How to practise this document

- Pass 1: cover the green boxes and reproduce them from memory.
- Pass 2: read only the ranked speed tricks and say when each shortcut is valid.
- Pass 3: do timed official questions and record whether errors were knowledge, method selection, algebra/arithmetic, or interpretation.
- Pass 4: revisit only the unit that generated the error; do not reread everything.